(1.5.1) *Proof.* Let $G = (V, E)$ be a graph without odd cycles; we show that G is bipartite. Clearly a graph is bipartite if all its components are bipartite or trivial, so we may assume that G is connected. Let T be a spanning tree in G, pick a root $r \in T$, and denote the associated tree-order on V by $\leqslant_T$. For each $v \in V$, the unique path rTv has odd or even length. This defines a bipartition of V; we show that G is bipartite with this partition.

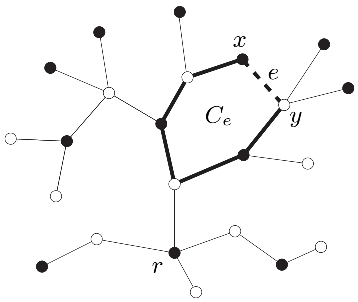

Fig. 1.6.3. The cycle C_e in $T + e$

Let $e = xy$ be an edge of G. If $e \in T$, with $x <_T y$ say, then $rTy = rTxy$ and so x and y lie in different partition classes. If $e \notin T$ then $C_e := xTy + e$ is a cycle (Fig. 1.6.3), and by the case treated already the vertices along xTy alternate between the two classes. Since C_e is even by assumption, x and y again lie in different classes. □

1.7 Contraction and minors

In Section 1.1 we saw two fundamental containment relations between graphs: the 'subgraph' relation, and the 'induced subgraph' relation. In this section we meet two more: the 'minor' relation, and the 'topological minor' relation. Let X be a fixed graph.

A *subdivision* of X is, informally, any graph obtained from X by 'subdividing' some or all of its edges by drawing new vertices on those edges. In other words, we replace some edges of X with new paths between their ends, so that none of these paths has an inner vertex in $V(X)$ or on another new path. When G is a subdivision of X, we also say that G *is a* TX.[7] The original vertices of X are the *branch vertices of the* TX; its new vertices are called *subdividing vertices*. Note that

subdivision TX of X

branch vertices

[7] The 'T' stands for 'topological'. Although, formally, TX denotes a whole class of graphs, the class of all subdivisions of X, it is customary to use the expression as indicated to refer to an arbitrary member of that class.

subdividing vertices have degree 2, while branch vertices retain their degree from X.

If a graph Y contains a TX as a subgraph, then X is a *topological minor* of Y (Fig. 1.7.1).

topological minor

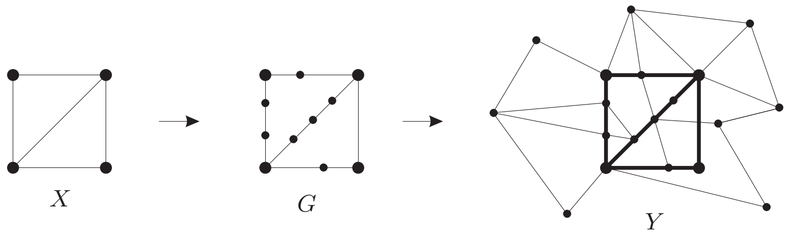

Fig. 1.7.1. The graph G is a TX, a *subdivision* of X. As $G \subseteq Y$, this makes X a *topological minor* of Y.

Similarly, replacing the vertices x of X with disjoint connected graphs G_x, and the edges xy of X with non-empty sets of G_x–G_y edges, yields a graph that we shall call an IX.[8] More formally, a graph G *is an* IX if its vertex set admits a partition $\{ V_x \mid x \in V(X) \}$ into connected subsets V_x such that distinct vertices $x, y \in X$ are adjacent in X if and only if G contains a V_x–V_y edge. The sets V_x are the *branch sets of the* IX. Conversely, we say that X arises from G by *contracting* the subgraphs G_x and call it a *contraction minor* of G.

IX

branch sets

contraction

If a graph Y contains an IX as a subgraph, then X is a *minor* of Y, the IX is a *model of* X in Y, and we write $X \preccurlyeq Y$ (Fig. 1.7.2).

minor, $\preccurlyeq$
model

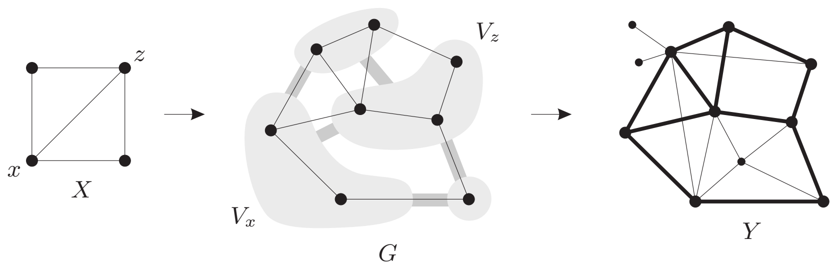

Fig. 1.7.2. The graph G is a model of X in Y, which makes X a *minor* of Y.

Thus, X is a minor of Y if and only if there is a map φ from a subset of $V(Y)$ onto $V(X)$ such that for every vertex $x \in X$ its inverse image $\varphi^{-1}(x)$ is connected in Y and for every edge $xx' \in X$ there is an

[8] The 'I' stands for 'inflated'. As before, while IX is formally a class of graphs, those admitting a vertex partition $\{ V_x \mid x \in V(X) \}$ as described below, we use the expression as indicated to refer to an arbitrary member of that class.

edge in Y between the branch sets $\varphi^{-1}(x)$ and $\varphi^{-1}(x')$ of its ends. If the domain of φ is all of $V(Y)$, and $xx' \in X$ whenever $x \neq x'$ and Y has an edge between $\varphi^{-1}(x)$ and $\varphi^{-1}(x')$ (so that Y is an IX), we call φ a
contraction *contraction* of Y onto X.

Since branch sets can be singletons, every subgraph of a graph is also its minor. In infinite graphs, branch sets are allowed to be infinite unless specified otherwise. For example, the graph shown in Figure 8.1.1 is an IX with X an infinite star.

[12.6.1] **Proposition 1.7.1.** *The minor relation $\preccurlyeq$ and the topological-minor relation are partial orderings on the class of finite graphs, i.e. they are reflexive, antisymmetric and transitive.* □

If G is an IX, then $P = \{ V_x \mid x \in X \}$ is a partition of $V(G)$, and we
G/P write $X =: G/P$ for this contraction minor of G. If $U = V_x$ is the only
G/U non-singleton branch set, we write $X =: G/U$, write v_U for the vertex
v_U $x \in X$ to which U contracts, and think of the rest of X as an induced subgraph of G. The 'smallest' non-trivial case of this is that U contains exactly two vertices forming an edge e, so that $U = e$. We then say that
contracting an edge $X = G/e$ arises from G by *contracting the edge* e; see Figure 1.7.3.

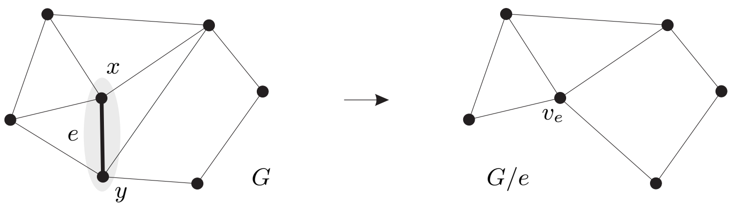

Fig. 1.7.3. Contracting the edge $e = xy$

Since the minor relation is transitive, every sequence of single vertex or edge deletions or contractions yields a minor. Conversely, every minor of a given finite graph can be obtained in this way:

Corollary 1.7.2. *Let X and Y be finite graphs. X is a minor of Y if and only if there are graphs $G_0, \ldots, G_n$ such that $G_0 = Y$ and $G_n = X$ and each G_{i+1} arises from G_i by deleting an edge, contracting an edge, or deleting a vertex.*

Proof. Induction on $|Y| + \|Y\|$. □

Finally, we have the following relationship between minors and topological minors:

Proposition 1.7.3. [4.4.2] [7.3.1] [12.7.3]

(i) *Every TX is also an IX (Fig. 1.7.4); thus, every topological minor of a graph is also its (ordinary) minor.*

(ii) *If $\Delta(X) \leqslant 3$, then every IX contains a TX; thus, every minor with maximum degree at most 3 of a graph is also its topological minor.* □

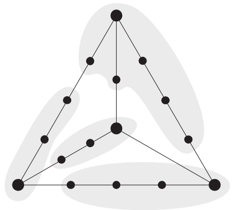

Fig. 1.7.4. A subdivision of K^4 viewed as an IK^4

Now that we have met all the standard relations between graphs, we can also define what it means to embed one graph in another. Basically, an *embedding* of G in H is an injective map $\varphi\colon V(G) \to V(H)$ that preserves the kind of structure we are interested in. Thus, φ embeds G in H 'as a subgraph' if it preserves the adjacency of vertices, and 'as an induced subgraph' if it preserves both adjacency and non-adjacency. If φ is defined on $E(G)$ as well as on $V(G)$ and maps the edges xy of G to independent paths in H between $\varphi(x)$ and $\varphi(y)$, it embeds G in H 'as a topological minor'. Similarly, an embedding φ of G in H 'as a minor' would be a map from $V(G)$ to disjoint connected vertex sets in H (rather than to single vertices) so that H has an edge between the sets $\varphi(x)$ and $\varphi(y)$ whenever xy is an edge of G. Further variants are possible; depending on the context, one may wish to define embeddings 'as a spanning subgraph', 'as an induced minor' and so on, in the obvious way. *embedding*

1.8 Euler tours

Any mathematician who happens to find himself in the East Prussian city of Königsberg (and in the 18th century) will lose no time to follow the great Leonhard Euler's example and inquire about a round trip through the old city that traverses each of the bridges shown in Figure 1.8.1 exactly once.

Thus inspired,[9] let us call a closed walk in a graph an *Euler tour* if it traverses every edge of the graph exactly once. A graph is *Eulerian* if it admits an Euler tour. *Eulerian*

[9] Anyone to whom such inspiration seems far-fetched, even after contemplating Figure 1.8.2, may seek consolation in the *multigraph* of Figure 1.10.1.